

Card Recommendation: Lander & Retro Jacket

Lander: Provides a stable platform for instruments and astronauts.

Cost: \$15 Million

Mass: 25kg

Power: 0W

Science Return: 5

Retro Rockets: Used to slow descent for controlled landings & works well with landers

Cost: \$10 million

Mass: 40kg

Power: 0W

Science Return: 5

I suggest only purchasing the lander and retro rockets. The lander provides a safe, flat platform for astronauts (if sent) and heavy instruments to deploy and operate. The retro rockets allow for a soft and controlled descent onto the lunar surface. After some research, this combo is proven, efficient, and consistent with NASA's Artemis programs strategies for crewed lunar exploration.

Why Not The Following Cards:

Airbags: Not ideal for crewed or fragile science payloads, they work better for small rovers or probes. There is a risk of bounce and crash

Hypersonic Parachute: No need since the moon lacks atmosphere and the hypersonic parachute is useful on atmospheric entries., it would be useful if we chose the Mars mission. We would need it if we return to Earth.

Maybe Card:

Heatshield:

Cost: \$5 Million

Mass: 20kg

Power: 0W

Science Return: 3

Yes if returning to Earth and if not, then no, since descent to the Moon doesn't require atmospheric protection.

Hypersonic Parachute:

Cost: \$8 Million

Mass: 20kg

Power: 0W

Science Return: 3

Total Mass:

65kg

Total Mass (Including Heat Shield & Hypersonic Parachute):

105kg

Total Cost:

\$25 Million

Total Cost (Including Heat Shield):

\$38 Million

Total Science Return:

10

Total Science Return (Including Heat Shield & Hypersonic Parachute):

16

Total Power:

0W

Total Power (Including Heat Shield & Hypersonic Parachute):

0W